a) A and D are correct b) B and D are correct
c) A,C and D are correct **d) A,B and C are correct**

Chapter 5 - Molecular Genetics

1. The association of histone H1 with a nucleosome indicates (NEET 2017)
   a) Transcription is occurring
   b) DNA replication is occurring
   c) The DNA is condensed into chromatin fibre
   d) The DNA double helix is exposed
2. Which of the following is not required for any of the techniques of DNA fingerprinting available at present? [NEET 2016]
   a) Zinc finger analysis
   b) Restriction enzymes
   c) DNA-DNA hybridization
   d) polymerase chain reaction
3. Satellite DNA is important because it [AIPMT 2015]
   a) codes for proteins needs in cell cycle
   b) shows high degree of polymorphism in population and also the same degree of polymorphism in an individual, which is heritable from parents to children
   c) Does not code for protein and is same in all members of the population.
   d) Codes for enzymes needed for DNA replication.
4. The diagram shows an important concept in the genetic implication of DNA. Fill in the blanks A to C. (NEET 2013)

   A B C

   DNA ----→mRNA ----------→protein -proposed by---
   a) A- transcription, B- replication C-James Watson
   b) A- transcription, B- transcription, C-Erwin
   c) A- transcription, B- translation, C-Francis Crick
   d) A- transcription, B- extension, C-Rosalind Frankin
5. Select the two statements out of the four (I –IV) given below about lac operon. [AIPMT 2010]
   i. Glucose or galactose may bind with the repressor and inactive it.
   ii. In the absence of lactose, the repressor binds with the operator region
   iii. The z-gene codes for permease.
   iv. This was elucidated by Francois Jacob and Jacques monod.
   The correct statements are
   a) i and ii b) i and iii **c) ii and iv** d) i and ii
6. Which one of the following pairs of codons is correctly matched with their function or the single for the particular amino acid? [AIPMT 2008]
   a) GUU, GCU – Alanine
   b) UAG, UGA – Stop codon
   c) AUG, ACG - start/methionine
   d) UUA, UCA – Leucine
7. The Okazaki fragments in DNA chain growth (AIPMT 2007)
   a) Result in transcription
   b) Polymerise in the 3' to 5' direction and forms replication fork
   c) Prove semi- conservative nature of DNA replication
   d) Polymerises in the 5' to 3'direction and explain 3' to 5' DNA replication
8. During translation initiation in prokaryotes, a GTP molecules is needed in [AIPMT 2003]
   a) association of 30s, mRNA with formyl met tRNA
   b) association of 50s subunit of ribosome with initiation complex
   c) formation of formyl met tRNA
   d) binding of 30s subunit of ribosome with mRNA.
9. Reverse transcriptase is
   a) RNA dependent RNA polymerase
   b) DNA dependent RNA polymerase
   c) DNA dependent DNA polymerase
   d) RNA dependent DNA polymerase
10. *Escherichia coli* fully labeled with N14 medium. The two strands of DNA molecules of the first generation bacteria have
    a) Different density and do not resemble parent DNA
    b) Different density but resemble parent DNA
    c) Same density and resemble parent DNA
    d) Same density but do not resemble parents DNA

Chapter 6 - Evolution

1. The wings of a bird and of an insect are
   a) homologous structure and represent convergent evolution
   b) homologous structure and represent divergent evolution
   c) analogous structure and represent convergent evolution
   d) analogous structure and represent divergent evolution
2. Which one of the following statement is correct?
   a) stem cells are specialized cells
   a) there is no evidence of the existence of gills during embryogenesis of mammals
   b) all plant and animal cells are totipotent
   c) Ontogeny repeats phylogeny
3. In Hardy-Weinberg equation, the frequency of heterozygous individual is represented by
   a) p^2 **b) 2pq** c) pq d) q^2
4. The correct order in Era is
   a) Palaeozoic---- Archaeozoic --- Coenozoic
   b) Archaeozoic ---Palaeozoic------ Proterozoic
   c) Palaeozoic--- Mesozoic ----- Coenozoic
   d) Mesozoic ---- Archaeozoic---- Proterozoic
5. The most apparent change during the evolutionary history of *Homo sapiens* is raced in (AIPMT 2010)
   a) loss of body hair
   b) walking upright
   c) shortening of jaws
   d) remarkable increase in the brain size
6. The process by which organisms with different evolutionary history evolve similar phenotypic adaptations in response to a common environmental challenge is called (AIPMT 2013)
   a) Natural selection
   b) Convergent evolution
   c) Non-random evolution
   d) Adaptive radiation Human health and diseases

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